1.4 Conversion of fractions to percentage and percentage to fractions

Conversion of fractions to percentage

Percentage means out of hundred $\left(\frac{x}{100}\right)\%$

Example 9.

Express $\frac{3}{8}$ as a percentage.

$\frac{3}{8}$ Out of hundred

$$= \frac{3}{\cancel{8}_{2}} \times \cancel{100}^{25}\% = \frac{75}{2}$$

$$= 35\frac{1}{2}\%$$

Exercise 5.

1. Convert these fractions to percentages. Show your working out.

a) $\frac{3}{3}$ b) $\frac{1}{3}$ c) $\frac{5}{6}$ d) $\frac{13}{20}$ e) $\frac{5}{8}$

2. Johana scored 13 out of 18 in Kiswahili test. What were his marks as percentage? Show your working out.

3. In a class of 45 pupils there are 18 girls. What percentage of the total number of pupils were boys? Show your working out.

4. Pamela had 60 hens. She sold 15 hens. What percentages of hens were unsold? Show your working out.

5. A basket had 36 fruits, 27 of them were ripe. What percentage of fruits was raw? Show your working out.

6. Abdi had 600 cattle. If he had 180 dairy cattle. What percentage were beef cattle?

7. In a tray there were 30 eggs. 11 eggs were rotten. What percentages of eggs were good?

Activity 3:

In groups discuss, where do we apply converting fractions to percentages.

Conversion of percentage to fractions

Example 10.

Write 35% as a fraction and write in simplest form.

35% is 35 out of 100

Change to a fraction $= \frac{35}{100}$

Simplify by cancelling the numerator and the denominator by a common divisor. $\frac{\cancel{35}}{\cancel{100}} = \frac{7}{20}$

$$= \frac{7}{20}$$

Example 11.

Express $33\frac{1}{3}\%$ as a fraction.

$$\frac{33\frac{1}{3}\times 3}{100\times 3} = \frac{100}{300}$$

$$= \frac{\cancel{100}}{\cancel{300}} = \frac{1}{3}$$

Multiply numerator and denominator by 3 to get whole number to get $\frac{100}{3\times 100} = \frac{100}{300}$

Then simplify by dividing 300 by 100 to get $\frac{1}{3}$

Exercise 6.

Convert these percentages to fractions in the simplest form.

a) 60% b) 75% c) 90% d) $32\frac{1}{2}\%$

e) $27\frac{1}{2}\%$ f) $2\frac{1}{4}\%$ g) $37\frac{1}{2}\%$ h) $66\frac{2}{3}\%$

Chose two questions tell your partner how you converted this percentages to fractions.

1.5 Conversion of decimals to percentage and percentage to decimals

Conversion of decimals to percentages

Example 12.

Express 0.05 as percentage.
Change to a fraction and multiply by 100%

$$0.05 = \frac{5}{\cancel{100}} \times \cancel{100}$$

$$= 5\%$$

Change it into fraction first i.e. $\frac{5}{100}$.

Then multiply by 100 and cancel 100 by 100 to get 5%

Activity 4:

In pairs, express the following as percentage.

a) 0.567 b) 0.4 c) 0.036 d) 0.48 e) 0.135

f) 1.75 g) 0.23 h) 2.8 i) 0.25 j) 3.75

Conversion of percentages to decimals

Example 13.

Convert 88% to a decimal

$88\% = \frac{88}{100} = 0.88$

Change it into fraction.

Then divide by 100

$= 0.88$

Exercise 7.

Convert the following percentages to decimals.

a) 77% b) 135% c) 265% d) 1%

e) 857% f) 13% g) 175% h) 8%

i) 19% j) 9%

Activity 5:

Where do we apply converting fractions to percentages?

1.6 Application of fractions, decimals and percentage

When we talk, we often use different words to express the same thing. For example, we could describe the same car as **tiny** or **little** or **small**. All of these words mean the car is not big.

Fractions, decimals, and percents are like the words tiny, little, and small. They're all just different ways of expressing parts of a whole.

Fractions
Fractions are used in the real world during jobs such as a chef or a baker because you need to know how much of something like butter or milk to put in a recipe

Decimals
Decimals are used in measurements for example my pen is 5.5 inches long.
Architects use decimals when they are measuring the height of a building.

Percent
In restaurant they have to use percent when they make a pizza so that they can cut it into equal pieces. Finally they use them when they decide how much of their budget goes to supplies.

Example 14.

In real life, fractions are used in games like soccer we talk of half time, as they are splint in to halves. Also fractions are used in food, i.e. $\frac{1}{2}$ cup of sugar.

In real life, percentages are used in liquids and food. i.e. 30% of tea is milk, 100% percent orange juice. Also in washing, we say what percentage of germs will be killed and how safe it is. i.e. 100% safe and kills 99.9% germs

Exercise 8.

In pairs, work out the following and share your working with your partner.

1. In a closing-down sale a shop offers 50% off the original prices. What fraction is taken off the prices?
2. In a survey one in five people said they preferred milk. What is this figure as a percentage?
3. Mary is working out a problem involving $\frac{1}{4}$. She needs to enter this into a calculator. How would she enter $\frac{1}{4}$ as a decimal on the calculator?
4. Deng pays tax at the rate of 25% of his income. What fraction of Deng's income is this?
5. When a carpenter was buying his timber, he had to put down a deposit of $\frac{1}{10}$ the value of timber. What percentage was this?
6. I bought my coat in January with $\frac{1}{3}$ off the original price. What percentage was taken off the price of the coat?
7. Brian bought a cloth that was 1.75 metres long. How could this be written as a fraction?